

RYAN HAR

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Mathematics graduate targeting quantitative trading and research. Placed **676** of **18,803** teams — top **3.6%**, solo against teams of up to **5** — in IMC Prosperity 4, running market-making, mean-reversion, pairs-trading, and options strategies across **5** live rounds. **170/170** GRE Quantitative, **800** SAT Math, **800** SAT Math II. Multiple **1st**-place finishes in international music competition and a Carnegie Hall performance.

AWARDS & HONORS

IMC Prosperity 4 — Global Algorithmic Trading Competition

Solo Competitor | Apr 2026

Market Making · Mean Reversion · Pairs Trading · Options

- Finished **676** of **18,803** teams (top **3.6%**) from over **100** countries, competing solo against teams of up to **5**.
- Researched historical tick data to find tradable edge across synthetic commodity and derivative products, screening candidate signals with regression, ACF, and stationarity tests before committing capital.
- Traded and retuned **4** strategy families across **5** live rounds — inventory-aware market-making, mean-reversion, pairs-trading, and dynamic options — re-fitting parameters between rounds as the regime shifted rather than defending a static book.

International Music Competitions — Cello

Multiple 1st-Place Finishes | 2011 – 2020

- Took **1st** place in multiple international competitions and placed in **11** competitions overall.
- Performed at Carnegie Hall and trained under internationally recognized teachers across **18+** years of daily practice.

EDUCATION & CERTIFICATIONS

Vanderbilt University — B.A., Double Major: Mathematics and Music

Nashville, TN | Aug 2020 – Dec 2024

GPA: **3.4/4.0**. Testing: GRE **170/170** Quantitative, **167** Verbal; SAT **800** Math, **730** Verbal; SAT Math II **800**.

Coursework: Probability & Statistics, Linear Algebra, Differential Equations, Complex Variables, Theory of Numbers, Mathematical Logic, Game Theory. Vanderbilt Academic Achievement Merit Scholarship.

Certifications

2024 – 2026

Baruch College CUNY — C++ Programming for Financial Engineering · Erasmus University Rotterdam — Econometrics: Methods and Applications · IBM Data Science Specialization (**12** courses).

QUANTITATIVE PROJECTS

Crypto Statistical Arbitrage

Python, pandas, statsmodels | 2026

- Built and stress-tested a dollar-neutral **cross-sectional crypto momentum** strategy using a 60-coin Binance.US dataset; incorporated **20 bps** transaction costs and **rolling walk-forward validation**, with realized BTC **beta below 0.1**
- Identified look-ahead bias in liquidity-based universe selection and replaced it with a point-in-time filter; attribution analysis showed out-of-sample performance was concentrated in a single ZEC momentum episode, limiting evidence of a persistent edge

Limit Order Book Matching Engine

C++, LOBSTER NASDAQ order flow | 2026

Execution Microstructure · Queue Priority · Slippage · Book Depth

- Reconstructed the venue side of execution by building a **matching engine** over real NASDAQ (LOBSTER) order flow — building mechanical intuition for how queue position sets fill priority, how crossing the spread pays slippage down the book, and how maker/taker flow (~93% add/cancel) builds and drains liquidity.
- Calibrated a synthetic order-flow model to published microstructure statistics — power-law limit-order depth ($\beta \approx 2.2$) and a **Geometric-Brownian-Motion** mid — to stress-test fill behavior and latency across p50/p99/p99.9 regimes.
- Implemented in C++**20** with cache-conscious, zero-allocation structures (hand-rolled AVL tree, open-addressing hash, slab pools) at 37M msg/s and malloc-free, bounded tail latency.

Option Pricing Comparison Tool

Python, NumPy, SciPy, Streamlit | Feb – Apr 2025

- Priced options by **2** independent methods — closed-form Black-Scholes and Monte Carlo — reconciled them for convergence, and benchmarked both against live market quotes to locate where theoretical and traded pricing diverge.

EXPERIENCE & LEADERSHIP

Summer Research Fellow — beAuthentic

Palo Alto, CA | Jun – Aug 2021

- Designed and ran behavioral experiments quantifying how **4** song attributes (tempo, key, instrumentation, lyrics) drove listener response; delivered the correlation analysis and visualizations that the company shipped into the product.
- Worked with the co-founder daily to tighten the data-acquisition pipeline; findings drove gains in user retention and engagement.

Co-Founder, Treasurer & Head Instructor — Vanderbilt Taekwondo Club

Nov 2021 – Dec 2024

- Co-founded the club from zero through university approval, recruited **30+** attendees to the first class, managed the budget as treasurer, and taught white through black belt as head instructor.

SKILLS & INTERESTS

- Quantitative & Trading:** Econometrics, regression analysis, statistical and mathematical modeling, time-series analysis (ACF, stationarity), probability, Monte Carlo simulation, options pricing (Black-Scholes), algorithmic and discretionary trading, market microstructure (queue priority, slippage, book depth).
- Programming:** Python (pandas, NumPy, SciPy), C++ (low-latency, cache-conscious design), Git.
- Interests:** Poker; Blackjack; Valorant (top **0.1%**); Cello performance; Taekwondo (Kukkiwon **2nd**-degree black belt); Dance.